



Alliance School of Advanced Computing
Alliance University – 562106

Departments of CSE, IT and Specializations

Process Manual

on

**Design Project (5CS1990 / 5IT1990, 6CS1991 /
6IT1991, E1CSA313 / E1ITA306)**

(Process, Guidelines, Assessment and Rubrics)

Prepared By

Dr. M. Selvam

Chief Coordinator / ASAC

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1. Design Project

1.1. Description

The Design Project is a culminating, often design, experience in an engineering curriculum where students apply theoretical knowledge, technical skills, and engineering principles to solve a real-world, open-ended problem. It typically spans one or two semesters and mimics a professional engineering environment. Students work individually or (more commonly) in teams to identify a need, define specifications, generate alternative solutions, analyze constraints (economic, environmental, social, ethical), build prototypes or models, test performance, and communicate results via reports and presentations.

Key characteristics:

- Open-ended problem statement (no single correct answer)
- Involves design, analysis, fabrication, and testing
- Follows a structured design process (e.g., need-finding, conceptual design, detailed design, prototyping, evaluation)
- Integrates multiple engineering disciplines or sub-disciplines

1.2. Significance

The Design Project is often called the “keystone” or “design” of engineering education because it:

- **Bridges theory and practice:** Connects classroom learning (math, science, core engineering courses) to real engineering challenges.
- **Prepares students for industry:** Simulates workplace demands such as teamwork, project management, client interactions, budget control, and iterative design.
- **Demonstrates competency:** Serves as a tangible portfolio piece for employers and graduate schools.
- **Fulfills accreditation requirements:** Required by bodies like ABET (Criterion 5: Design) to ensure graduates can design systems, components, or processes under realistic constraints.
- **Fosters professional identity:** Transitions students from learners to practicing engineers by making them responsible for decisions, failures, and successes.

1.3. Objectives

The main educational objectives of the Design Project are to enable students to:

1. **Apply the engineering design thinking process** from problem definition to final solution.
2. **Incorporate multiple realistic constraints** (e.g., cost, safety, sustainability, manufacturability, ethics).
3. **Integrate knowledge** from mathematics, basic sciences, and engineering sciences.



4. **Function effectively in a team** – including roles, conflict resolution, and collaborative decision-making.
5. **Communicate technical information** clearly through written reports, oral presentations, and visual aids.
6. **Manage a project** using tools like Gantt charts, budgets, and design reviews.
7. **Prototype and test** a solution, analyzing data to refine the design.
8. **Consider broader impacts** – societal, environmental, and global contexts.

1.4. Outcomes

After successfully completing a design project, students should be able to:

Cognitive Outcomes:

- Formulate a clear problem statement and design requirements from vague client needs.
- Generate and evaluate multiple alternative solutions using decision matrices.
- Perform engineering analysis (e.g., stress, thermal, circuit, or fluid analysis) to support design choices.
- Select materials, components, and manufacturing processes appropriately.

Practical/Technical Outcomes:

- Build a functional prototype or simulation model.
- Use engineering software in the design process.
- Conduct experiments or tests to validate performance against specifications.

Professional/Soft Outcomes:

- Produce a comprehensive final design report and deliver a professional presentation.
- Demonstrate effective teamwork (peer evaluations, meeting logs, division of labour).
- Manage project timeline and budget within allocated resources.
- Reflect on ethical, sustainability, and societal implications of the design.

1.5. Complex Problems (Characteristics)

Engineering design projects inherently involve **complex engineering problems** as defined by accreditation bodies (e.g., ABET). These problems possess most or all the following attributes:

Attribute	Example in Design Project
Deep technical knowledge required	Cannot be solved by simple textbook formula; requires synthesis of multiple engineering topics (e.g., mechanical + electrical + control systems).



Attribute	Example in Design Project
Open-ended, multiple solutions	No single “correct answer”; trade-offs exist between cost, performance, safety, and aesthetics.
Involves conflicting constraints	Lightweight vs. strong, low-cost vs. durable, fast vs. energy efficient.
Not fully defined in advance	Requirements evolve as the team discovers technical or customer constraints.
Requires information gathering	Literature review, vendor data, standards (ISO, ASME, IEEE), user feedback.
Non-standard solution path	Cannot be solved by routine application of a standard method; requires creativity, iteration, and judgment.
Involves real-world constraints	Budget limits, manufacturing tolerances, safety regulations, supply chain issues, environmental impact.
Requires teamwork/coordination	Different sub-systems must interface correctly; integration challenges arise.

Examples of complex problems from engineering design projects:

- Designing an autonomous waste-sorting robot for a campus cafeteria (mechanical + electrical + programming + user safety + budget).
- Developing a low-cost water filtration system for a rural community (fluid mechanics + materials + social acceptance + testing constraints).
- Creating a portable ECG monitor (biomedical + electronics + power management + FDA-like safety considerations).

1.6 Curricular Requirements

In our curriculum, it is incorporated into two phases

1. Design Project -1 (5CS1990) in Vth Semester which covers Analysis, Design and Implementation
2. Design Project – 2 (6Cs1991) in VIth Semester which covers the full-fledged extended Implementation and Deployment



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It is expected to have the following outcomes

1. Design Project -1 with Implementation
2. Design Project -2 with Extended Implementation, Deployment

This will be supported by possible publications in terms of publication in Scopus Indexed IEEE International Conferences. The other possible outcomes may be with well-defined process, prototype, product, patent, etc.



2.0 Design Thinking

Unlike traditional linear engineering design (waterfall model), Design Thinking is **human-centered, iterative, and empathetic**. It is particularly powerful for tackling the *complex or ill-defined problems* typical of design projects.

2.1 Design Thinking in Engineering Projects

Design Thinking is a non-linear, iterative process that seeks to understand the user, challenge assumptions, and redefine problems before attempting to find solutions. When applied to an engineering design project, it shifts the focus from "building a technical solution" to "solving a human need."

The five phases are:

- **Empathize** – Understand the user's experience and emotions.
- **Define** – Frame the core problem based on user insights.
- **Ideate** – Generate a wide range of possible solutions.
- **Prototype** – Build low-fidelity, inexpensive versions of the solution.
- **Test** – Gather user feedback and refine.

Unlike traditional design (requirements → design → build → test), Design Thinking loops back and forth between phases.

2.2 Significance of Design Thinking in Engineering Projects

Traditional Engineering Approach	Design Thinking Approach
Problem given by instructor/client	Problem discovered through user observation
Requirements fixed early	Requirements evolve with empathy
Solution judged by technical specs	Solution judged by user satisfaction & usability
Failure is undesirable	Failure is a learning opportunity (fast iteration)
Linear progression	Iterative loops



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Significant outcomes will be

- **Reduces mismatch** between engineering solution and actual user need.
- **Encourages creativity** beyond obvious technical fixes.
- **Manages complexity** by breaking large problems into user-centered insights.
- **Prepares students for human-centered industries** (healthcare, consumer products, assistive technology, smart systems).
- **Aligns with ABET outcomes** – especially those related to understanding stakeholder needs, societal context, and iterative testing.

2.3 Objectives of Using Design Thinking in the Project

When implementing a design project using Design Thinking, the objectives are to enable students to:

1. **Empathize with end-users** through interviews, observations, or immersion.
2. **Define a meaningful problem statement** (point of view) rather than accepting a given one.
3. **Generate diverse solution concepts** without early judgment (divergent thinking).
4. **Build quick, low-cost prototypes** (paper, cardboard, role-play, or basic electronics) to test ideas rapidly.
5. **Fail early and often** to learn quickly and improve the final design.
6. **Iterate based on user feedback**, not just technical validation.
7. **Integrate technical analysis** within an empathic, user-driven framework.

2.4 Outcomes from the student point of view

After implementing a design project with Design Thinking, students should be able to:

Empathy & User Research Outcomes:

- Conduct effective user interviews and translate observations into actionable insights.
- Create user personas, journey maps, and empathy maps.

Problem Definition Outcomes:

- Write a “How Might We...” statement that captures the real user need.
- Distinguish between symptomatic problems and root causes.

Ideation Outcomes:

- Use techniques like brainstorming, SCAMPER, or analogies to generate 50+ ideas.
- Select promising concepts via user-aligned criteria (not just technical feasibility).

Prototyping Outcomes:

- Build storyboards, wireframes, or physical low-fidelity prototypes in hours/days, not weeks.
- Use prototyping to test specific questions



Testing & Iteration Outcomes:

- Capture user feedback systematically and prioritize changes.
- Demonstrate at least two full iteration cycles based on user input.

Engineering Integration Outcomes:

- Translate a validated low-fidelity prototype into a high-fidelity engineering prototype (CAD, circuits, code).
- Document both the human-centered journey and the technical decisions.

2.5 Addressing Complex Problems in Engineering Projects

Complex Problem Attribute	How Design Thinking Helps
Ill-defined / vague	Empathy & Define phases uncover the real need; reframing turns ambiguity into actionable problem statements.
Deep technical knowledge needed	Ideation generates both technical and non-technical concepts; later phases filter for technical feasibility.
Open-ended / many solutions	Divergent-convergent pattern explores wide solution space before narrowing.
Conflicting constraints	Prototyping tests trade-offs (e.g., cost vs. usability) with real users, not just theory.
Requires information gathering	User feedback becomes a primary data source, complementing technical literature.
Non-standard solution path	Iteration encourages exploring unexpected paths after each test.
Real-world constraints (cost, safety, ethics)	Prototypes reveal constraints early; testing uncovers ethical or safety issues users might raise.
Requires teamwork	Design Thinking workshops (e.g., affinity mapping, journey mapping) foster collaboration and shared mental models.



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2.6 Challenges & Mitigations

Challenge	Mitigation
Students trained in linear thinking resist iteration	Use graded milestones for each iteration loop; show examples of famous engineering failures due to lack of empathy.
Time pressure in one-semester design project	Start with low-fidelity prototypes in first 4 weeks; technical refinement later.
Difficulty finding real users	Partner with campus clinics, local NGOs, or use role-playing (students as surrogate users).
Overemphasis on empathy → neglect of technical rigor	Assign separate grades for “user validation” and “engineering analysis”; require both.
Team conflict over iterative changes	Use design sprint techniques (e.g., 5-day sprints) with clear decision rules.



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3. Process

3.1 Identification of the Problem Statement

The first and most critical step in any engineering design project is the identification of a clear, relevant, and well-defined problem statement. This stage involves recognizing a genuine need or an opportunity for improvement within a real-world context, often derived from observations, stakeholder interviews, industrial challenges, or societal gaps such as inefficiency, safety hazards, accessibility issues, or environmental concerns. The problem statement must be specific, measurable, and neither too narrow (which limits creativity) nor too broad (which makes the project unmanageable), and it typically emerges from an empathetic understanding of end-users combined with preliminary fact-finding. A well-articulated problem statement—for example, "Design a low-cost, solar-powered water purification system for rural households with no grid electricity"—serves as the North Star for all subsequent design decisions, preventing scope creep and ensuring that the final solution addresses a legitimate, high-impact need rather than an imagined or trivial one.

3.2 Defining the Scope, Objectives, Outcomes, Commercial Value and Social Impact

Once the problem statement is established, the project team must rigorously define the scope, objectives, outcomes, commercial value, and societal impact to transform a vague idea into a feasible engineering plan. The scope sets clear boundaries—what will and will not be included—such as specific performance metrics, budget limits, time constraints, and resource availability, thereby preventing the project from expanding uncontrollably. Objectives are specific, measurable, achievable, relevant, and time-bound (SMART) goals, such as "achieve 99% bacterial removal at a flow rate of 2 liters per hour," while outcomes describe the tangible deliverables like a working prototype, technical drawings, simulation results, or a user manual. Equally important is articulating the commercial value—how the solution reduces cost, saves time, improves quality, or creates a marketable product—and the broader impact, including environmental benefits (e.g., reduced plastic waste), social benefits (e.g., improved health in underserved communities), or ethical advantages (e.g., fair labour in manufacturing). This holistic framing ensures that the project is not merely a technical exercise but a viable, responsible, and value-driven engineering intervention.

3.3 Literature Review

The literature review is a systematic exploration of existing scientific papers, patents, product catalogs, technical standards, and case studies to understand what has already been done in the problem domain and to identify knowledge gaps that the current project can fill. This stage prevents reinventing the wheel by revealing prior solutions, their successes, and their limitations, while also providing benchmark data for performance comparison, design rules, material properties, and safety regulations. A thorough literature review typically covers academic databases (IEEE, Scopus, Google Scholar), industry standards (ISO, ASTM), commercial products, and even failure analysis reports to learn from past mistakes. Beyond informing technical decisions, the literature review helps refine the problem statement, justify design choices, avoid patent infringement, and establish a theoretical foundation for the project, which becomes a critical section of the final term paper. For engineering



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students, this process also builds essential skills in information retrieval, critical analysis, and citation management, transforming them from passive learners into active researchers who can position their work within the broader engineering landscape.

3.4 Analysis & Design

The analysis and design phase translates conceptual ideas into detailed, quantifiable, and verifiable engineering specifications through mathematical modelling, simulations, and iterative refinement. Analysis involves calculating forces, heat transfer, circuit behaviour, fluid dynamics, or control system responses using theoretical equations, finite element analysis (FEA), computational fluid dynamics (CFD), or circuit simulators (SPICE) to predict how the proposed design will perform under various conditions. Design then takes those analytical results and creates concrete artifacts such as CAD models, schematic diagrams, bill of materials (BOM), and engineering drawings that specify dimensions, tolerances, materials, and manufacturing processes. This phase is highly iterative: analysis may reveal a stress concentration that requires geometry changes, or a thermal simulation might indicate the need for a heat sink, leading back to redesign. The outcome is a design that is not only creative but also mathematically justified, meeting all safety factors, performance targets, and regulatory requirements while balancing trade-offs between cost, weight, speed, and durability.

3.5 Methodologies / Technologies

Selecting appropriate methodologies and technologies involves choosing the development framework, project management approach, and specific technical tools that will guide the execution of the design project from concept to prototype. Methodologies can include Agile (for software-heavy projects), Waterfall (for well-defined sequential tasks), Design Thinking (for user-centered, iterative development), or hybrid models like the V-model for safety-critical systems, each offering different advantages in terms of flexibility, documentation, or risk management. The technologies section lists the hardware (sensors, microcontrollers, actuators, materials), software (CAD, simulation, compilers, version control), and fabrication methods (3D printing, CNC machining, PCB etching) that will be employed, along with justifications for each choice based on performance, cost, availability, and team expertise. Importantly, this phase also considers technology readiness levels (TRL), supply chain reliability, and interoperability between subsystems, ensuring that the selected stack is not only powerful but also practical to implement within the project's time and budget constraints.

3.6 Module Description

The module description breaks the overall design project into smaller, manageable, and functionally independent subsystems or modules, each with a clear purpose, input-output relationship, and interface definition. For example, an autonomous delivery robot might have modules for power management (battery, voltage regulation), locomotion (motors, wheels, encoders), perception (camera, LiDAR, ultrasonic sensors), navigation (microcontroller, SLAM algorithm), and communication (WiFi/Bluetooth module). Each module is described in terms of its internal function, the technologies it uses, its performance specifications, and crucially, how it interfaces with other modules—pin assignments, data protocols (I2C, SPI, UART), mechanical mounting points, and



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power budgets. This modular approach enables parallel work by different team members, simplifies testing and debugging (each module can be tested independently), facilitates reuse of designs in future projects, and makes the overall system easier to document, understand, and troubleshoot. A well-structured module description also highlights potential integration risks, such as timing conflicts or electromagnetic interference, which can be addressed before final assembly.

3.7 Implementation

The implementation phase is where the detailed design from previous stages is physically constructed, coded, assembled, and integrated into a functional prototype or system that operates in a real or simulated environment. This stage involves procuring components, fabricating parts (via 3D printing, machining, PCB etching, or laser cutting), writing and debugging software, assembling subsystems, and performing integration testing to ensure that all modules communicate and function together as intended. Implementation is rarely linear; engineers encounter unexpected issues such as parts that do not fit, sensors that give noisy readings, timing conflicts between modules, or power supply instability, requiring on-the-fly redesign, rework, and creative problem-solving. A disciplined approach includes version control for both hardware (revision tracking of CAD files and schematics) and software (Git), daily build logs, and incremental integration—adding one module at a time and verifying its operation before adding the next. Successful implementation produces a working prototype that may be a proof-of-concept (demonstrating feasibility), a breadboard model (for testing), or a pre-production unit (nearing final form), and it often requires iterative cycles of build-test-fix until the system meets the predefined performance specifications.

3.8 Results and Discussion

The results and discussion section presents the quantitative and qualitative data collected from testing the implemented prototype, followed by a critical interpretation of what those findings mean in relation to the original objectives and the broader engineering context. Results are typically shown using tables, graphs, oscilloscope captures, thermal images, or statistical summaries, comparing measured performance (e.g., power consumption, response time, accuracy, efficiency) against the design targets established in earlier phases. Discussion goes beyond simply restating numbers; it explains why certain results were achieved—for example, a lower-than-expected flow rate might be traced to pump cavitation, or a slower control loop response might be due to interrupt conflicts in firmware. This section also compares the project's outcomes with those from the literature review, highlighting improvements, trade-offs, or limitations relative to existing solutions. Crucially, the discussion acknowledges anomalies, errors, and unexpected behaviors, analyzing their root causes (measurement error, environmental factors, design flaws) and assessing whether the prototype is fit for its intended purpose. A strong discussion convinces the reader that the team understands not only what happened but also the underlying engineering principles, and it provides evidence-based answers to whether the project successfully solved the original problem statement.



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3.9 Conclusion and Future Directions

The conclusion concisely summarizes the entire project—restating the problem, the approach taken, the major results achieved, and the key lessons learned—without introducing new data or analysis, serving as a final takeaway for the reader. It answers three essential questions: Did we achieve our objectives? What are the most significant contributions of this work? And what are the practical implications of our findings? Following the conclusion, the future directions section outlines specific, actionable recommendations for extending or improving the project beyond its current scope, acknowledging that no engineering design is ever truly complete. These may include technical enhancements (e.g., using a more efficient motor, implementing wireless data logging, reducing weight through advanced materials), additional testing (e.g., long-term reliability studies, user acceptance trials, extreme environment validation), scaling strategies (e.g., moving from manual assembly to automated manufacturing, designing for low-cost mass production), or new applications of the same core technology. Good future directions are realistic and prioritized, often identifying the single most important next step (e.g., "Before field deployment, the waterproofing must be upgraded to IP67 standards") rather than an unattainable wish list. Together, the conclusion and future directions demonstrate engineering maturity: celebrating what was achieved while humbly recognizing limitations and charting a path for continuous improvement.

3.10 Report Writing and Publishing the Term Paper

The final stage is the comprehensive documentation and dissemination of the entire project through a structured term paper, technical report, and potentially a conference or journal publication. Report writing follows a standard engineering format—abstract, introduction, problem statement, literature review, methodology, design details, results, discussion, conclusion, and references—and must include all technical data, schematics, code listings, test results, and photographs of the prototype. Beyond mere description, a good term paper critically analyzes what worked, what failed, why certain design decisions were made, and what lessons were learned, including an honest discussion of limitations and future improvements. Publishing the term paper, whether in an institutional repository, a student journal, or a peer-reviewed conference, adds academic credibility, allows other engineers to build upon the work, and provides students with a tangible portfolio piece that demonstrates their ability to communicate complex technical ideas effectively. For many engineering programs, this report also serves as the primary assessment artifact, graded not only on technical correctness but also on clarity, organization, adherence to formatting guidelines, and the quality of visual aids—making report writing an essential professional skill as important as the design itself.



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4.0 Administration

4.1 Orientation Session on Design Thinking and Project

Students attend a mandatory workshop introducing Design Thinking phases (Empathize, Define, Ideate, Prototype, Test) and the project's lifecycle, timelines, and evaluation criteria. They learn how human-centered methodology differs from traditional linear engineering and why it is critical for solving complex, ill-defined problems. Students also receive handouts on documentation expectations, ethical guidelines, and intellectual property basics.

4.2 Formation of Student Batches

Students self-form or are assigned into batches of 4 members, ensuring interdisciplinary diversity and balanced workload distribution. A Faculty Handler is allotted to a set of batches who is responsible for communication, task allocation, and deadline tracking. Batches are finalized within one week of orientation, and any grouping conflicts are resolved by the faculty handler.

4.3 Collection of Project Description from Students

Each student batch submits a one-page preliminary project proposal including a tentative title, problem statement, real-world need, and expected outcomes. The proposal also identifies broad domain areas and any initial constraints or resources required. Submissions are collected digitally through a centralized learning management system with a strict cutoff deadline.

4.4 Collection of Faculty Members' Area of Interest and Expertise

Faculty members submit a standardized form listing their technical domains, research interests, prior industry experience, and preferred project types (e.g., hardware-heavy, simulation-based, software-focused). This information is compiled into a searchable mentor database that also indicates each faculty member's maximum mentoring capacity (number of batches). The database is updated annually to reflect new faculty skills and ongoing research directions.

4.5 Mapping the Area & Expertise and Allocation of Mentors

A project coordinator matches each student batch with a faculty mentor whose expertise aligns with the batch's proposed domain, using a systematic mapping matrix that considers technical fit, workload balance, and student preferences. In cases of domain mismatch or high demand, batches may be asked to refine their proposals or are assigned to a mentor with adjacent expertise. Final mentor allocation is published within two weeks, and students are given three days to appeal the allocation with valid justification.



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4.6 Orientation Session for Faculty Members on the Roll Out

Faculty mentors and handlers attend a dedicated session covering their roles, responsibilities, evaluation rubrics, review schedules, and the expected frequency of student interactions. They receive guidelines on how to guide without over-directing, how to assess Design Thinking artifacts (personas, journey maps, prototypes), and how to document student progress. The session also clarifies academic integrity policies, conflict resolution procedures, and the format for submitting mentor feedback after each review.

4.7 Maintenance of Review Diary by the Students

Each student batch maintains a physical or digital review diary that chronologically records meeting dates, discussion points, decisions made, action items, and reflective comments on challenges and learnings. The diary must be signed or timestamped by the faculty mentor after every interaction and is presented during all project reviews as evidence of consistent progress. Students are graded on diary completeness, regularity of entries, and the quality of reflections (not just task completion).

4.8 Interaction between Students and Faculty Members

Students are required to meet with their assigned mentor at least once every two weeks, either in person or via a recorded virtual meeting, to present progress, seek clarification, and receive formative feedback. Mentors maintain office hours specifically for project batches and may call additional sessions if milestones are at risk. All interactions are briefly documented in the review diary, and any unresolved technical or team conflicts are escalated to the program coordinator within three days.

4.9 Allotment of Reviewers for Project Reviews I, II, III and End Semester Examinations

A review committee allocates independent reviewers (different from the primary mentor) for each batch across three internal reviews and the final end-semester viva voce, ensuring impartial assessment and fresh perspectives. Reviewers are chosen based on domain relevance, availability, and rotation across batches to avoid bias. The allotment schedule is published at the start of the semester, and each review panel consists of at least two faculty members, with external industry experts invited for the final examination when possible.

4.10 Submission of Project Reports

Students submit a structured term paper (typically 40–60 pages) following a prescribed template that includes all sections from problem identification through conclusion and future directions, along with appendices containing code, schematics, and test data. The report undergoes a plagiarism check using institutional software, and submissions are accepted only through the learning management system by the announced deadline. Late submissions incur a predefined penalty (e.g., 5% marks deduction per day), and no submissions are accepted after the end-semester examination schedule is published.

4.11 End Semester Examination: Viva Voce

Students defend their design project before a panel of internal and external examiners through a 20-minute oral examination that includes a live demonstration of the working prototype and a question-answer session probing technical depth, design decisions, and understanding of limitations. The viva voce assesses individual contributions (each team member is questioned separately), ability to articulate design thinking rationale, and response to constructive criticism. Marks are awarded based on prototype functionality, clarity of presentation, depth of knowledge, and the student's ability to propose valid future directions.



5. Evaluation and Rubrics

5.1 Components of Design Project - I

- Review -1 will be conducted and evaluated on Problem identification, need and scope, Literature review and gap analysis, Objectives and requirements, Proposed methodology, tools and feasibility and Presentation and technical defence.
- Review -2 will be conducted and evaluated on System architecture, Module description, Implementation progress — target 50% of the approved scope, Presentation and viva and Project demonstration
- Review -3 will be conducted and evaluated on Implementation completeness, Demonstration, Usability and user satisfaction, Submission of the paper and Report quality and conformance
- Final Viva Voce examination will be conducted and evaluated on the Project Report submitted, Problem Formulation, Review of Literature, Design Methodology, and Concept of the Work / Originality and Significance & Relevance – Application in the Real World and Implementation in terms of Presentation, Demonstration and Viva Voce

5.2 Components of Design Project - II

- Review -1 will be conducted and evaluated on Consolidation of Semester V work and closure of feedback, Refined scope, objectives and extended design, Implementation plan and early implementation evidence, Presentation and technical defence and Project demonstration.
- Review -2 will be conducted and evaluated on System architecture and integration design, Module development and technical quality, Implementation progress — target 70% of the extended scope, Presentation and viva and Project demonstration
- Review -3 will be conducted and evaluated on Implementation completeness — 100% of the extended scope, Demonstration, Testing, validation and results against baseline, Submission of the paper and Report quality and conformance
- Final Viva Voce examination will be conducted and evaluated on the Project Report submitted, Problem Formulation, Review of Literature, Design Methodology, and Concept of the Work / Originality and Significance & Relevance – Application in the Real World and Implementation in terms of Presentation, Demonstration and Viva Voce

5.3 Parameters of Evaluation

Parameters of Evaluation in Reviews I, II, III and End Semester Examination (Viva Voce) are appropriately selected based on the expected components of the Reviews and Examinations described above.



5.4 DSA Components

In Design Project – I, the components are

- Review -1 – 10 Marks (Converted from Mentor Marks 25 + Reviewer marks 25)
- Review -2 – 10 Marks (Converted from Mentor Marks 25 + Reviewer marks 25)
- Review -3 – 10 Marks (Converted from Mentor Marks 25 + Reviewer marks 25)
- Project Report with Publication Details – 10 Marks & Assessment by Faculty Handler – 10 Marks
- Mentor Marks – 20 Marks
- Attendance (Only for Eligibility)
- Viva Voce Examination – 30 Marks

In Design Project – II, the components are

- Review -1 – 10 Marks (Converted from Mentor Marks 25 + Reviewer marks 25)
- Review -2 – 10 Marks (Converted from Mentor Marks 25 + Reviewer marks 25)
- Review -3 – 10 Marks (Converted from Mentor Marks 25 + Reviewer marks 25)
- Project Report with Publication Details – 10 Marks & Assessment by Faculty Handler – 10 Marks
- Mentor Marks – 20 Marks
- Attendance (Only for Eligibility)
- Viva Voce Examination – 30 Marks

5.5 Rubrics on Literature Review, Presentation and Term Paper / Report

Evaluation Rubrics

Criteria	Excellent (5)	Good (4)	Satisfactor y (3)	Needs Improvement (2)	Unsatisfacto ry (1)
Understa nding of Design Thinking	Proposes creative solutions to improve existing design thinking approaches.	Applies design thinking framework to analyse real-world challenges.	Demonstrates clear understanding of design thinking and its impact in engineering.	Explains key concepts of design thinking but lacks in relating with Engineering applications.	Unable to relate design thinking with Engineering aspects for complex problem solving



User Research Methods	Develops innovative user research techniques specifically tailored to engineering problems.	Analyses user research data to identify user needs and pain points with critical insights.	Effectively utilizes various user research methods (interviews, observations , empathy mapping) to gather accurate data.	Can describe basic methods but struggles to apply them.	Vague understanding of user research methods.
Empathy Building	Champions user-centred design and advocates for empathetic problem-solving in engineering.	Uses empathy to generate impactful design solutions that truly address user needs.	Demonstrates strong empathy for users and can articulate their needs and frustrations clearly.	Attempts to build empathy but lacks in collaboration.	Shows limited engagement with user perspectives.
Creativity and Collaboration	Leads the team in exploring unconventional approaches and fostering innovation culture in engineering.	Drives creative thinking within the team and fosters an open and supportive environment .	Actively participates in ideation sessions and encourages collaboration.	Follows instructions but lacks initiative.	Shows limited contribution to team discussions.



Ideation Techniques:	Develops and shares novel ideation techniques applicable to complex engineering challenges.	Adapts and combines ideation techniques effectively to produce diverse and original solutions.	Implements various ideation techniques (brainstorming, sketching, role-playing) with some flexibility.	Relies on basic brainstorming techniques.	Struggles to generate creative solutions.
Research Article	Problem definition and research article is prepared but the components of the research articles are eligible to published and plagiarism level is less than 10%	Problem definition and research article is prepared but the components of the research articles are eligible to present but the plagiarism level is more than 15%	Problem definition and research article is prepared but the components of the research articles are not eligible to present	Problem is designed but not converted to research article	Have not written any Research article and not even defined problems



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6. Guidelines and Evaluation Rubrics in Detail

6.1 Guidelines

1. Projects must be approved by the Mentor before starting of the process.
2. All works must be original; plagiarism is strictly prohibited.
3. Students can work in teams.
4. Teams must have 4 members.
5. Roles and contributions of each team member must be clearly defined.
6. A team leader must be designated for communication and coordination.
7. Team members must contribute fairly and equally.
8. Conflicts must be reported promptly to the Faculty Handler / Department Coordinator / Chief Coordinator
9. Students must meet regularly with their Mentor and Faculty Handler of the class weekly.
10. Meeting notes or progress reports must be maintained in terms of Mentor / Reviewer Diary
11. Follow official timelines for proposals, drafts, final reports, and presentations.
12. Maintain the documentation phase by phase.
13. Projects must follow ethical guidelines (no falsified data, respect IP, consent/privacy for surveys).
14. Follow the formatting guidelines (APA, IEEE, etc.) for all submissions.
15. Submit all required documents: proposal, report, logbook, and presentation materials.
16. Deliver a final presentation and be prepared for questions.
17. Demonstration of project with Coverage, Completeness and Convergence is mandatory
18. Maintain professional conduct with Mentor, Faculty Handler, Teammates, and evaluators.
19. Use institutional resources responsibly and uphold academic integrity.
20. Papers must be published in Scopus-Indexed IEEE International Conferences / reputed Journals in both semesters.



6.2 DSA Components and their Rubrics

Component	Mentor	Panel	Faculty Handler	Final Marks
Review–I	25	25	–	10
Review–II	25	25	–	10
Review–III	25	25	–	10
Report, publication and Faculty Handler assessment	10	–	10	20
Mentor — Continuous supervision	20	–	–	20
Total (Continuous Internal Evaluation)				70
Final Review — Semester-End Viva Voce Examination				30
Total				100

(Rubrics for DSA Components are attached in Annexure)

Annexures

1. Rubrics for DSA Components
2. Review Diary
3. Project Report Format
4. Template - Mentor Marks on Reviews
5. Template - Evaluation Sheet for Reviewers
6. Template - Report Marks by Mentors
7. Template - Faculty Handler Marks
8. Template - Mentor Marks on Continuous Supervision